

Transistors and MOSFETs

Section 21

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1 Section 21 — Transistors

Every circuit element met so far — the resistor, the sources, the capacitor, the inductor and, most recently, the diode — is a *two-terminal* device. This section first shows what a *third* terminal buys us, using the humble switch; the transistor is then introduced as exactly such a device, and the rest of the section develops its most important member, the MOSFET.

1.1 From Two-Terminal to Three-Terminal Devices

Two-terminal devices have a fixed I-V

The I-V characteristic of a two-terminal device is *fixed*:

- **Linear devices** — the voltage source, the resistor and the current source each present a fixed straight-line I-V.
- **Non-linear devices** — the diode presents a fixed *curve*, or its piecewise linear (PWL) approximation.

Nothing in the surrounding circuit can change *which* characteristic the device presents.

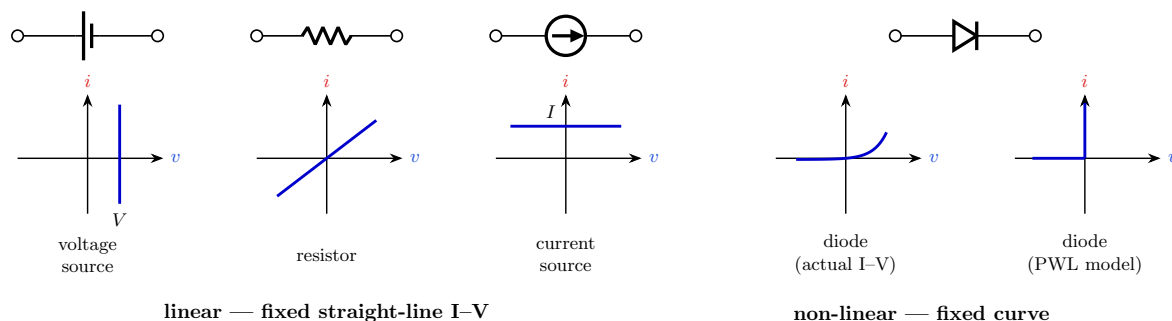


Figure 1: Two-terminal devices and their fixed I-V characteristics.

A **three-terminal device** removes this limitation: the I-V characteristic seen between two of its terminals, T_1 and T_2 , is *selected* by a control signal C applied at the third terminal. The simplest three-terminal device is the **switch**.

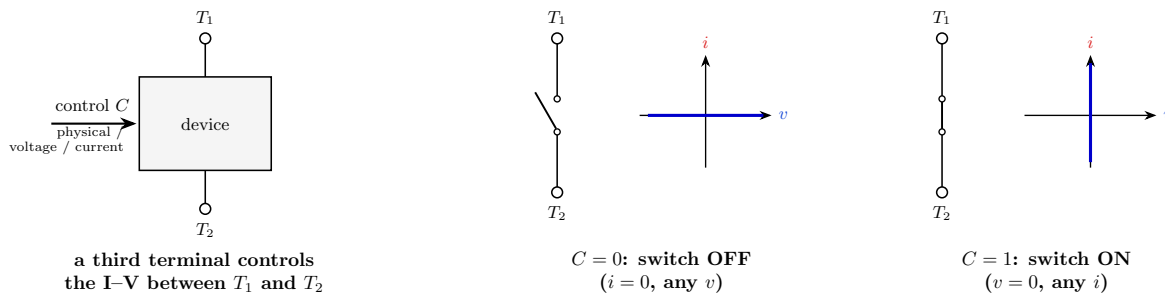


Figure 2: The switch as a three-terminal device. The control C chooses which of the two I-V characteristics appears between T_1 and T_2 : with $C = 0$ the switch is an open circuit; with $C = 1$ it is a short circuit.

control $C \rightarrow$ selects the I-V characteristic presented between T_1 and T_2

1.2 Electronic Switches

Depending on the control, a switch can be:

1. **Analog** — controlled by a physical toggle or button.
2. **Digital** — controlled by an electrical signal: a *voltage* or a *current*. Electrically controlled switches are precisely what *transistors* are — they are introduced next.

Binary representation. Digital circuits use two states (0/False, 1/True), so a circuit variable must be mapped onto two states of a device or parameter. The usual voltage convention is adopted throughout these notes:

$$\underbrace{5 \text{ V} \rightarrow 1}_{\text{HIGH}} \qquad \underbrace{0 \text{ V} \rightarrow 0}_{\text{LOW}}$$

1.3 Logic Gates from Switches

Switches wired to a bulb already compute logic:

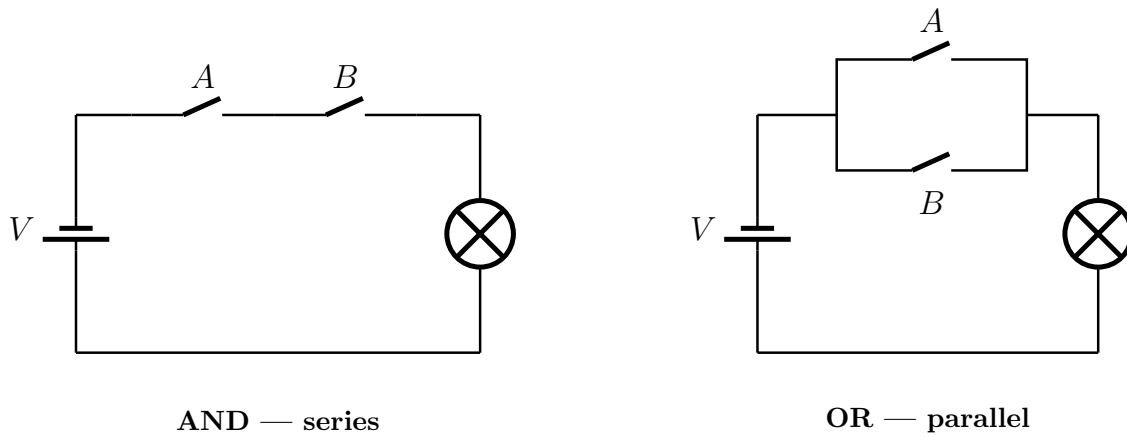


Figure 3: Switch logic. In series, the bulb lights only if *A* **and** *B* are closed; in parallel, it lights if *A* **or** *B* is closed.

Series switches $\rightarrow$ AND	Parallel switches $\rightarrow$ OR
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Bulb circuits cannot drive further gates. The *preferred* form replaces the bulb with a **pull-up resistor** and takes a *voltage* as the output, so one gate's output can drive the next gate's input (cascading):

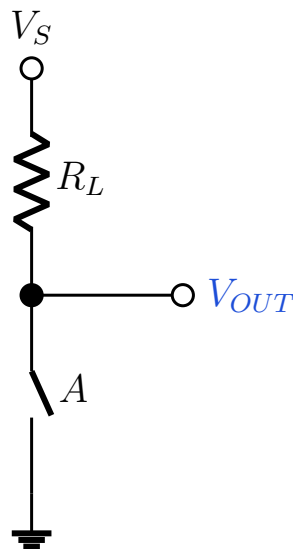


Figure 4: The pull-up form, shown for the NOT gate. Switch open ($A = 0$): no current flows, $V_{OUT} = V_S$ (logic 1). Switch closed ($A = 1$): the output is pulled to ground, $V_{OUT} = 0$ (logic 0).

The pull-up rule

switch network conducts $\Rightarrow V_{OUT} = 0$

switch network open $\Rightarrow V_{OUT} = V_S$

Because the switch network pulls the output *low* when it conducts, this topology naturally produces the *inverting* gates: NOT, NAND (series switches) and NOR (parallel switches).

1.3.1 Example 1 — Implementing a Logic Function with Switches**Problem**

Implement the logic function $f = (A + B)C$ using switches and a bulb.

The bulb must light when $(A \text{ or } B)$ **and** C are true. OR is a parallel pair and AND is a series connection, so the network is the parallel pair $A \parallel B$ in series with C :

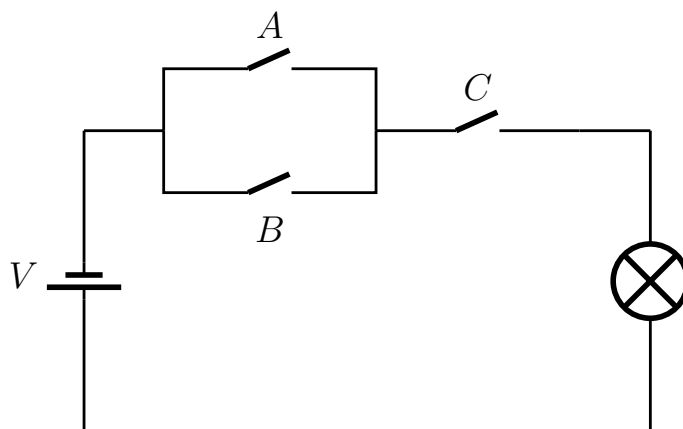


Figure 5: Switch implementation of $f = (A + B)C$.

1.4 Transistors**Definition**

Transistors are *three-terminal* semiconductor devices in which one terminal — the *control lead* — sets the response between the other two. They act as either **electrically controlled switches** or **amplifier controls**. A small voltage and/or current applied at a *control lead* regulates a much larger current flow through the other two leads — the way the small effort of turning a valve regulates a large flow of water.

Transistors appear in almost every electronic circuit. For example:

- Switching circuits

- Amplifier circuits
- Oscillator circuits
- Current-source and voltage-regulator circuits
- Power-supply circuits
- Digital logic ICs

Small control signal $\rightarrow$ transistor $\rightarrow$ large controlled current

1.5 Classification of Transistors

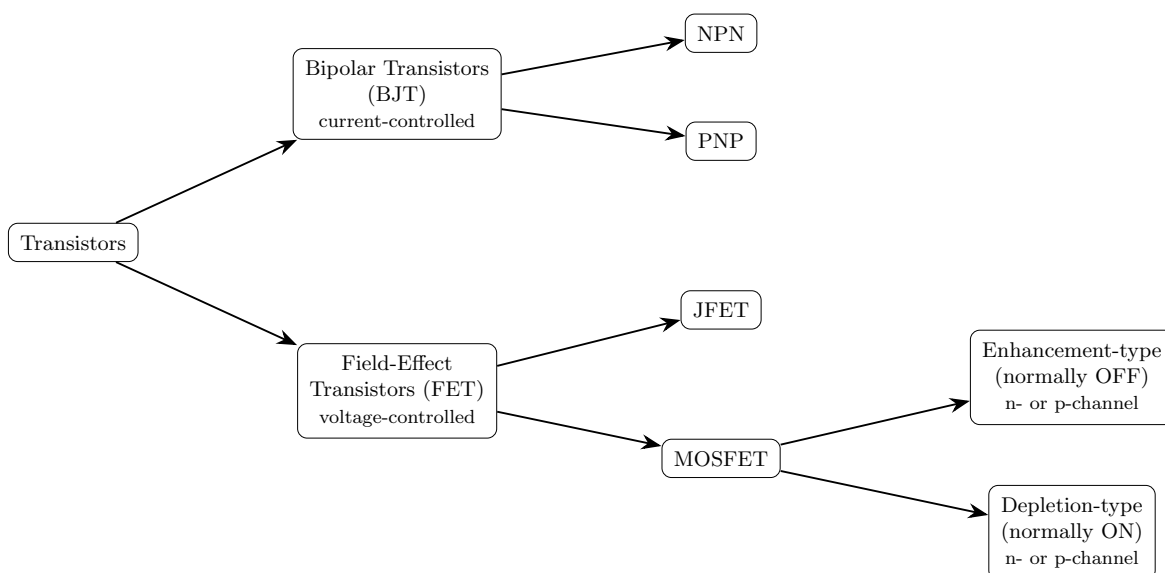


Figure 6: The transistor family. The two major families differ in *what* controls them: BJTs are controlled by a current, FETs by a voltage. These notes cover the MOSFET.

Table 1: The two major transistor families compared.

	Bipolar (BJT)	Field-Effect (FET)
Control input	biasing <i>current</i>	<i>voltage</i> only ($\approx$ no current)
Charge carriers used	both electrons and holes	one carrier type
Input impedance	moderate	very high ($\sim 10^{14} \Omega$)
Loading of the control circuit	draws a small current	essentially none

FETs dominate modern circuit design:

- They draw essentially zero current at the control lead.
- They are easier and cheaper to manufacture (less silicon).
- They can be made extremely small — which makes them the building block of integrated circuits.

- Drawback: lower transconductance than a BJT at the same current, hence lower voltage gain in simple amplifier circuits.

“Normally ON” and “normally OFF”

Normally refers to the state of the transistor when the control lead is left unbiased (for a MOSFET: gate shorted to source, $V_{GS} = 0$). A *normally OFF* device conducts no current in that state; a *normally ON* device conducts its maximum current. “ON” is not literal — the current is a *variable* quantity set by the magnitude of the control voltage.

1.6 The MOSFET

Definition

The **Metal Oxide Semiconductor Field-Effect Transistor (MOSFET)** is the most popular transistor in use today. A small voltage applied at its **gate** lead alters the current flow through its **drain–source** channel. A metal-oxide *insulator* between the gate and the channel gives the gate an extremely large input impedance ($\geq 10^{14} \Omega$, versus $\sim 10^9 \Omega$ for a JFET) — the gate draws essentially *no current at all*.

Its three terminals map directly onto the three-terminal picture that opened this section: the **gate** is the control terminal, and the controlled path runs from **drain** to **source**.

MOSFETs draw almost no input current, are easy to make, can be made extremely small, and consume very little power. They are used in ultrahigh-input-impedance amplifiers, voltage-controlled “resistor” circuits, switching circuits, and large-scale digital ICs.

1.6.1 Kinds of MOSFETs

1. **Enhancement-type** — *normally OFF* at $V_{GS} = 0$. Applying a gate-source voltage makes the channel *less* resistive.
2. **Depletion-type** — *normally ON* at $V_{GS} = 0$. Applying a gate-source voltage makes the channel *more* resistive.

Each kind comes in **n-channel** and **p-channel** form, giving four combinations:

Table 2: Effect of the gate-source voltage on the drain-source channel.

Type	Channel	Effect of V_{GS}
Depletion	n-channel	negative V_{GS} ($V_G < V_S$) $\rightarrow$ channel <i>more</i> resistive
Depletion	p-channel	positive V_{GS} ($V_G > V_S$) $\rightarrow$ channel <i>more</i> resistive
Enhancement	n-channel	positive V_{GS} ($V_G > V_S$) $\rightarrow$ channel <i>less</i> resistive
Enhancement	p-channel	negative V_{GS} ($V_G < V_S$) $\rightarrow$ channel <i>less</i> resistive

The rest of these notes develop the theory for the **n-channel enhancement-type** MOSFET (NMOS). *Detailed sections on the depletion-type and enhancement-type devices (characteristics, formulas and worked examples) will be added separately.*

Practical notes

- **Body terminal** — many MOSFETs bring out a fourth lead connected to the substrate. It forms a junction with the channel and must be held at a non-conducting voltage, usually by tying it to the source. Biasing it away from the source shifts the threshold voltage.
- **Electrostatic fragility** — the gate oxide is so thin (gate capacitance of only a few pF) that static charge picked up by handling can puncture it and destroy the device. Eliminate static from the work area before installing MOSFETs.

1.7 The MOSFET as a Switch — The S Model

As noted above, the theory from here onward is developed for the **enhancement-type** MOSFET — the device is normally *OFF*, and $V_{GS,th} > 0$ is the gate-source voltage at which it turns *ON*. The *S* model, and every equation in the rest of this section, is specific to this type.

The simplest model of the MOSFET is a *voltage-controlled switch* connected between drain and source, with the gate-source voltage as the control:

$$C = V_{GS} : \quad C = "0" \Rightarrow V_{GS} < V_{GS,th}, \quad C = "1" \Rightarrow V_{GS} \geq V_{GS,th}$$

where $V_{GS,th}$ is the **threshold voltage** of the MOSFET.

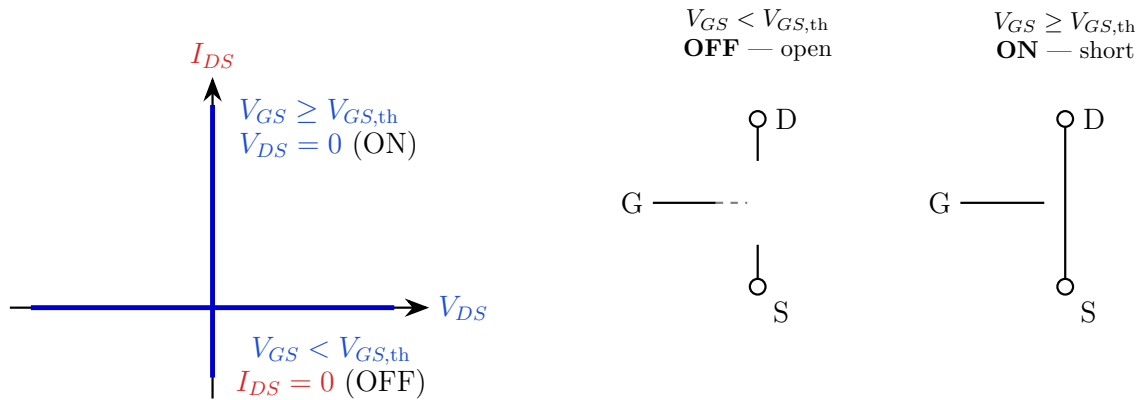


Figure 7: The S (switch) model. Left: the two-branch I-V characteristic between drain and source. Right: the equivalent circuit in each state.

S model (enhancement-type MOSFET)

$$\underbrace{V_{GS} < V_{GS,th} : I_{DS} = 0}_{\text{OFF} \Rightarrow \text{open circuit}}$$

$$\underbrace{V_{GS} \geq V_{GS,th} : V_{DS} = 0}_{\text{ON} \Rightarrow \text{short circuit}}$$

1.8 Logic Gates with MOSFETs

To build real logic gates, simply *replace the switches with MOSFETs* in the pull-up topology. The input drives the gate; the MOSFET conducts when its input is HIGH ($> V_{GS,th}$) and pulls the output LOW.

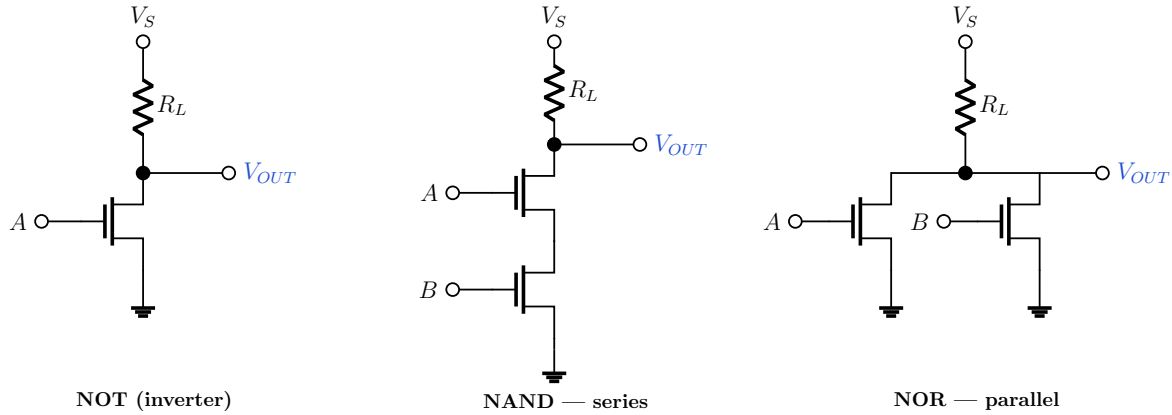


Figure 8: NMOS logic gates. Series MOSFETs pull the output low only when *all* inputs are high (NAND); parallel MOSFETs pull it low when *any* input is high (NOR).

Input HIGH ($V_{IN} > V_{GS,th}$) $\rightarrow$ MOSFET ON $\rightarrow$ output pulled LOW

1.9 Voltage Transfer Characteristic of the Inverter

The **voltage transfer characteristic (VTC)** plots the output voltage (y -axis) against the input voltage (x -axis). It is the tool used to check that a gate obeys a required *static discipline*.

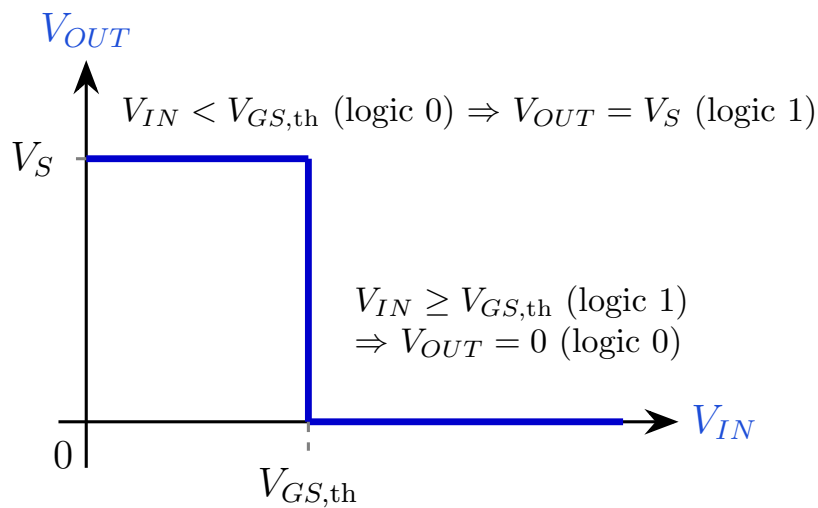


Figure 9: VTC of the NMOS inverter under the S model.

Inverter VTC

$$V_{IN} < V_{GS,th} \text{ (logic 0)} \Rightarrow V_{OUT} = V_S \text{ (logic 1)}$$

$$V_{IN} \geq V_{GS,th} \text{ (logic 1)} \Rightarrow V_{OUT} = 0 \text{ (logic 0)}$$

VTC with two inputs. A NAND or NOR gate has two inputs but the VTC has only one x -axis. The solution is to draw *two* VTCs — one with the other input fixed at 0, one with it fixed at 1. For the NAND gate: with $V_A = 0$ the output stays at V_S regardless of V_B ; with $V_A = 1$ the gate behaves exactly like an inverter in V_B .

1.10 Constructing a Real MOSFET

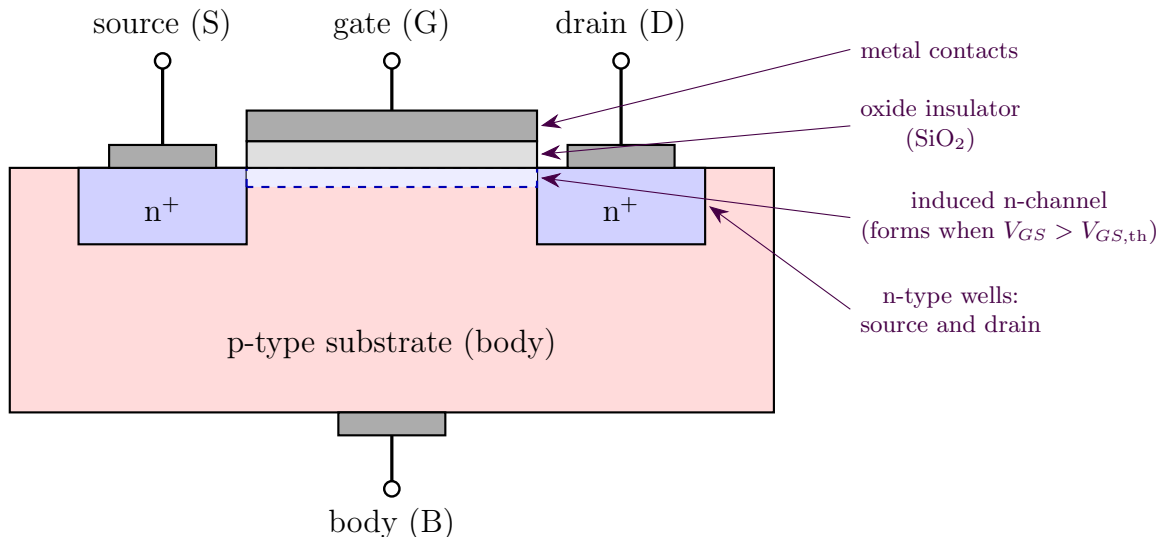


Figure 10: Physical structure of the n-channel enhancement-type MOSFET (NMOS): two n-type wells (source and drain) in a p-type substrate, an oxide insulator over the gap between them, and a metal gate on top.

1.10.1 How the Channel Forms

1. **Equilibrium** — left alone, electrons from the n-type wells diffuse into the p-type material to fill holes, leaving positive ions in the wells and negative ions in the substrate (exactly as at a *pn* junction).
2. **Small $V_{GS} > 0$** — holes are pushed away from the region under the gate; the device still cannot conduct. This is **cutoff**.
3. **$V_{GS} > V_{GS,th}$** — the attraction of the gate becomes strong enough that *free electrons collect under the oxide*. This **induced n-type channel** now connects the source well to the drain well.

4. **Apply** $V_{DS} > 0$ — electrons flow from source to drain through the channel; conventional current I_D flows from drain to source.

$V_{GS} > V_{GS,th}$ forms the channel $\rightarrow V_{DS}$ propels the current through it

1.10.2 Circuit Symbol and I–V Characteristic

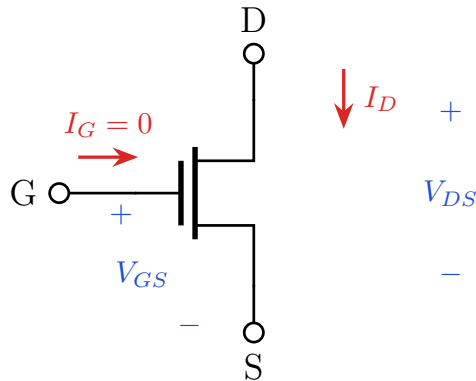


Figure 11: NMOS circuit symbol and sign convention. Because of the oxide insulator, the gate current is exactly zero.

Reading the terminals

- $I_G = 0$ — the insulated gate draws no current.
- A MOSFET is a three-terminal device, so there is *no single* I–V characteristic. The characteristic is defined as I_D versus V_{DS} for a *fixed* V_{GS} , and the curve changes as V_{GS} changes — a *family* of curves.

1.11 Modes of Operation

The definitions below are for the enhancement-type MOSFET, which is normally OFF.

For $V_{GS} \leq V_{GS,th}$ no channel exists and no current flows: **cutoff** mode. When $V_{GS} > V_{GS,th}$, the mode of current flow depends on how the *propelling* voltage V_{DS} compares with the *channel-inducing* voltage $V_{GS} - V_{GS,th}$:

- $V_{DS} < V_{GS} - V_{GS,th}$ — current is starting to flow and I_D rises rapidly with V_{DS} : the **ohmic** region.
- $V_{DS} \geq V_{GS} - V_{GS,th}$ — current has reached its maximum and barely increases with V_{DS} : the **active** region.

1.11.1 What the Channel Actually Does

The two regions are easiest to understand by watching the induced channel itself as V_{DS} is raised, holding $V_{GS} > V_{GS,th}$ fixed (Fig. 12):

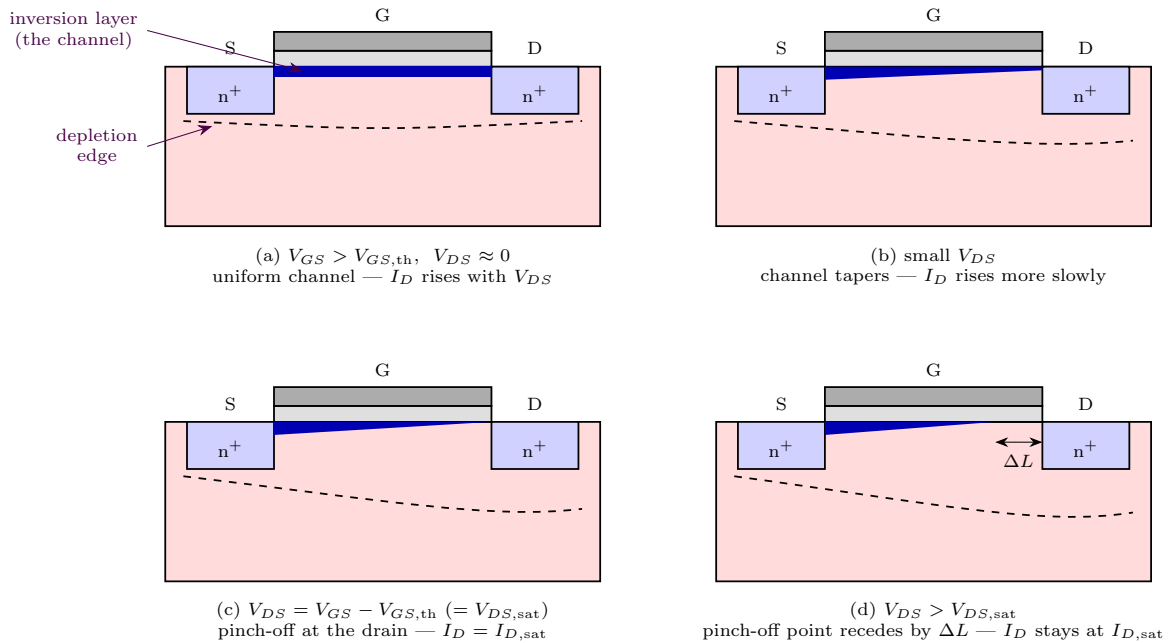


Figure 12: The induced channel as V_{DS} is raised at fixed $V_{GS} > V_{GS,th}$: uniform channel, tapered channel, pinch-off exactly at the drain, and the pinch-off point receding by ΔL .

1. $V_{DS} \approx 0$ — the channel (inversion layer) has uniform thickness from source to drain and behaves as a simple resistor: I_D rises in proportion to V_{DS} .
2. V_{DS} **small but growing** — the drain end of the channel sits at a higher potential than the source end, so the gate-to-channel voltage is *smaller* at the drain end, and the channel *tapers*. I_D still increases with V_{DS} , but at a decreasing rate.
3. $V_{DS} = V_{GS} - V_{GS,th}$: **pinch-off** — the gate-to-drain voltage has fallen to exactly the threshold, so the channel thickness at the drain end shrinks to *zero*. The current reaches its maximum value, $I_{D,sat}$.
4. V_{DS} **beyond pinch-off** — the pinch-off point moves slightly toward the source (by ΔL); the extra voltage is dropped across the pinched-off stretch rather than across the channel, so I_D does not increase further. This is the *active* region.

Pinch-off in these notes' symbols: $V_{DS,sat} = V_{OV}$

The lecture calls the boundary value of V_{DS} the **saturation voltage** $V_{DS,sat}$, and the current there $I_{D,sat}$. Pinch-off happens when the gate-to-drain voltage falls to the threshold, $V_{GS} - V_{DS} = V_{GS,th}$, i.e.

$$V_{DS,sat} = V_{GS} - V_{GS,th}$$

— the *overdrive voltage* V_{OV} of the NMOS-equations subsection, and exactly the ohmic/active

boundary used throughout these notes. $I_{D,\text{sat}}$ is the value the active-region current equation evaluates to.

A note on terminology

These notes name the two conducting regions **ohmic** and **active**, following the textbook. Other sources — the slides included — call the very same regions *triode* and *saturation*:

These notes	Also called (slides)	Behaviour
Ohmic region	triode mode	acts like a variable resistor
Active region	saturation mode	acts like a V_{GS} -controlled current source

The name *ohmic* emphasises the resistor-like behaviour near the origin; the name *triode* comes from the shape of the characteristic curve. Likewise, *active* emphasises that this is the region used for amplification, while *saturation* emphasises that the current has levelled off with respect to V_{DS} .

Similarly, the threshold voltage is written $V_{GS,\text{th}}$ throughout these notes, following the textbook; the slides write V_T for the same quantity — the gate-source voltage at which the enhancement-type device just begins to conduct.

Watch out — “saturation” means the opposite thing for a BJT

Sources that call the MOSFET’s *active* (constant-current) region “saturation” collide with bipolar-transistor language: for a BJT, “saturation” means fully ON — a closed switch. The two uses are unrelated, which is precisely why these notes (and the textbook) stick to *active region* for the MOSFET.

1.12 The Water Tap Analogy

The three modes map exactly onto a kitchen water tap:

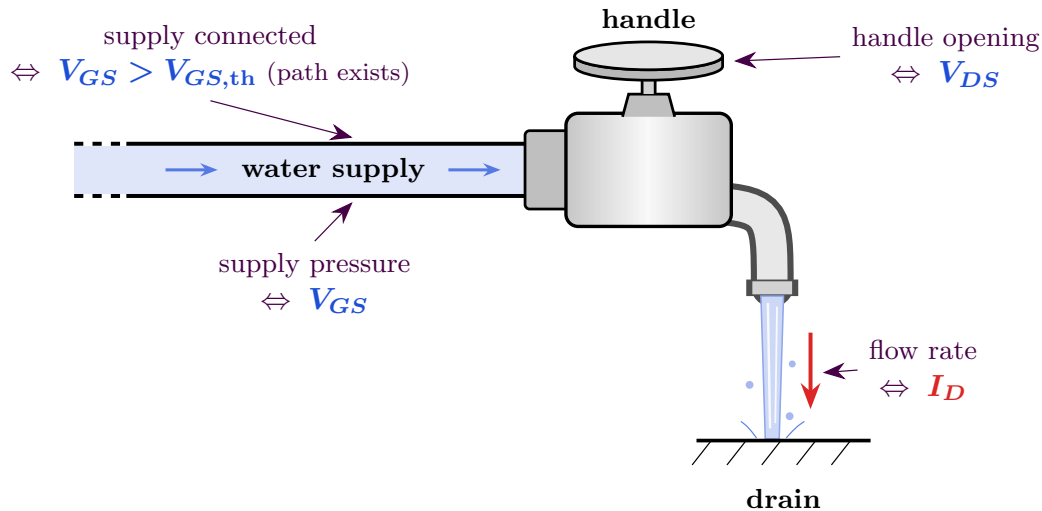


Figure 13: The water tap analogy. The *gate* voltage plays the role of the water supply; the *drain-source* voltage plays the role of the handle.

- **Connecting the water supply** $\Leftrightarrow$ setting V_{GS} above $V_{GS,th}$. It *establishes the path* for flow — without it, no amount of handle-turning produces water. **Cutoff = supply disconnected.**
- **A higher-pressure supply** $\Leftrightarrow$ a larger V_{GS} . More flow can *potentially* occur.
- **The handle** $\Leftrightarrow V_{DS}$. It controls the *actual amount* of flow.
- **Ohmic** = handle in its normal range: opening it further increases the flow (I_D rises with V_{DS}).
- **Active** = handle turned to (or past) the point of maximum flow: turning it further gains nothing — the flow is now limited by the supply pressure (I_D is set by V_{GS} , not V_{DS}).

Table 3: The water tap analogy, term by term.

Water tap	MOSFET
Supply connected (path exists)	$V_{GS} > V_{GS,th}$ (channel formed)
Supply disconnected	cutoff ($V_{GS} < V_{GS,th}$, $I_D = 0$)
Supply pressure	V_{GS}
Handle opening	V_{DS}
Water flow rate	I_D
Handle in normal range	ohmic (I_D increases with V_{DS})
Handle at/past maximum flow	active (I_D set by V_{GS})

The point of the analogy

The gate decides whether a path exists and how much current is *available*; the drain-source voltage decides how much of it *actually flows* — up to that limit. The two controls play different roles, exactly like the supply and the handle of a tap.

1.13 NMOS Equations (Enhancement-Type)**Scope of these equations**

Every equation in this subsection — the mode conditions, the ohmic-region current and the active-region current — describes the **enhancement-type** NMOS specifically. The depletion-type MOSFET is normally ON and is governed by a different pair of equations (I_{DSS} and $V_{GS,off}$ based); a first look at these comes with the textbook problems at the end of these notes, and the full treatment will follow in its own section.

Definitions

V_{GS}	gate-source voltage
V_{DS}	drain-source voltage
$V_{GS,th}$	threshold voltage — minimum gate voltage that forms the channel
$V_{OV} = V_{GS} - V_{GS,th}$	overdrive voltage — excess gate voltage after forming the channel
k'_n	process transconductance parameter ($= \mu_n C_{ox}$)
W/L	width-to-length (aspect) ratio of the channel
$k = \frac{1}{2} k'_n \frac{W}{L}$	MOSFET construction parameter

The three modes of the enhancement-type NMOS

Cutoff: $V_{GS} < V_{GS,th}$ $I_D = 0$

Ohmic: $V_{GS} \geq V_{GS,th}$, $V_{DS} < V_{OV}$ $I_D = k [2(V_{GS} - V_{GS,th}) - V_{DS}] V_{DS}$

Active: $V_{GS} > V_{GS,th}$, $V_{DS} \geq V_{OV}$ $I_D = k (V_{GS} - V_{GS,th})^2 = k V_{OV}^2$

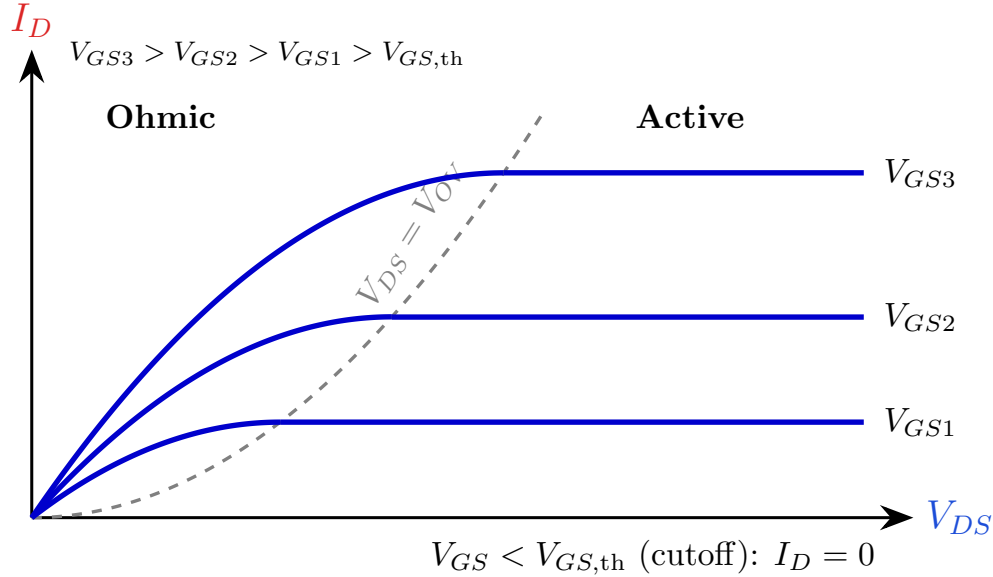


Figure 14: Output characteristics of the **n-channel enhancement-type** MOSFET (NMOS): a family of I_D - V_{DS} curves, one per V_{GS} . The dashed parabola $V_{DS} = V_{OV}$ separates the ohmic region from the active region. Note that all curves start at the origin — the device is *normally OFF*, so $I_D = 0$ until V_{GS} exceeds $V_{GS,th}$. The depletion-type device, being normally ON, has a different family of curves.

1.13.1 The Ohmic Region

- Near the origin the device behaves like a **voltage-controlled resistor**: I_D rises steeply with V_{DS} , at a rate set by V_{GS} . This resistor-like behaviour is what gives the region its name.
- The average voltage across the channel is $V_{OV} - \frac{1}{2}V_{DS}$; regrouping the ohmic equation as $I_D = 2k \left(V_{OV} - \frac{1}{2}V_{DS} \right) V_{DS}$ shows exactly this term at work.
- $I_D \propto k$ (and hence $\propto k'_n$ and W/L), $I_D \propto \left(V_{OV} - \frac{1}{2}V_{DS} \right)$.

1.13.2 The Active Region

- The average voltage across the channel is *fixed* at $\frac{1}{2}V_{OV}$, so the current no longer depends on V_{DS} .
- The device behaves as a **V_{GS} -controlled current source**: I_D is set entirely by the overdrive voltage, $I_D = k V_{OV}^2$.
- Because I_D is controlled by V_{GS} and is almost independent of V_{DS} , this is the region used for *amplification* — hence the name *active*.

$$\begin{aligned} V_{DS} < V_{OV} &\rightarrow \text{ohmic: resistor-like} \\ V_{DS} \geq V_{OV} &\rightarrow \text{active: current-source-like} \end{aligned}$$

2 Solving Circuits with MOSFETs

The mode of a MOSFET is not known before the circuit is solved — exactly the difficulty met earlier with the diode. The same remedy applies, now with *three* candidate states instead of two.

The Method of Assumed States

1. **Assume** — assume one of the modes: Cutoff, Ohmic or Active.
2. **Solve** — use the corresponding current equation together with KCL and KVL.
3. **Verify** — check whether the conditions on V_{GS} and V_{DS} are satisfied. If not, repeat with another mode.

Assume a mode $\rightarrow$ Solve $\rightarrow$ Verify $\rightarrow$ (if it fails, assume again)

A note on quadratics

Because the active and ohmic equations are *quadratic*, solving usually means solving $ax^2+bx+c=0$. This produces **two roots**. Choose the one that is favourable to (i.e. consistent with) your assumption — the other root always violates the mode conditions and must be discarded.

The conditions to verify are exactly those of Table 4 restated for convenience:

Table 4: Verification conditions for each MOSFET mode (enhancement-type).

Mode	Conditions	Current
Cutoff	$V_{GS} < V_{GS,\text{th}}$	$I_D = 0$
Ohmic	$V_{GS} \geq V_{GS,\text{th}}, V_{DS} < V_{OV}$	$I_D = k(2V_{OV} - V_{DS})V_{DS}$
Active	$V_{GS} > V_{GS,\text{th}}, V_{DS} \geq V_{OV}$	$I_D = kV_{OV}^2$

2.1 Example 2 — Assumption Rejected, Then Corrected

Problem

Analyse the circuit below to find I_D and V_{o2} using the Method of Assumed States. The input of the MOSFET is $V_{o1} = 1$ V, and the device has $V_{GS,th} = 0.2$ V and $k = 2$ mA/V². *You must validate your assumptions.*

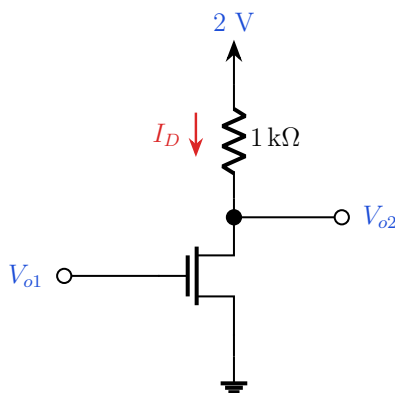


Figure 15: Circuit for Example 2. The source is grounded, so $V_{GS} = V_{o1}$ and $V_{DS} = V_{o2}$.

Since the source is grounded, $V_{GS} = V_{o1} = 1$ V and $V_{DS} = V_{o2}$, so the overdrive voltage is

$$V_{OV} = V_{GS} - V_{GS,th} = 1 - 0.2 = 0.8 \text{ V}$$

First Attempt — active

Assume: Let the MOSFET be in the **active region**, which requires

$$V_{GS} > V_{GS,th} \quad \text{and} \quad V_{DS} \geq V_{OV}, \text{ i.e. } V_{o2} \geq 0.8 \text{ V}$$

Solve: Using the active-region current equation and then Ohm's law on the 1 kΩ drain resistor:

$$\begin{aligned} I_D &= k (V_{GS} - V_{GS,th})^2 = 2 \times (0.8)^2 = 1.28 \text{ mA} \\ V_{o2} &= 2 - I_D \times 1 \text{ k}\Omega = 2 - 1.28 = 0.72 \text{ V} \end{aligned}$$

Verify: The active region requires $V_{o2} \geq 0.8$ V, but here $V_{o2} = 0.72 \text{ V} < 0.8 \text{ V}$.

Assumption is INCORRECT!

The device must be in the ohmic region.

Second Attempt — ohmic

Assume: Let the MOSFET be in the **ohmic region**, which requires $V_{DS} < V_{OV} = 0.8 \text{ V}$.

Solve: The ohmic equation and the resistor equation must give the same current:

$$I_D = k(2V_{OV} - V_{DS})V_{DS} \quad \text{and} \quad I_D = \frac{2 - V_{o2}}{1}$$

Equating them with $V_{DS} = V_{o2}$,

$$2(1.6 - V_{o2})V_{o2} = 2 - V_{o2}$$

which rearranges to the quadratic

$$V_{o2}^2 - 2.1V_{o2} + 1 = 0 \quad \implies \quad V_{o2} = 1.37 \text{ V} \text{ or } 0.73 \text{ V}$$

Only the root consistent with the assumption survives: the ohmic region requires $V_{DS} < 0.8 \text{ V}$, so $V_{o2} = 1.37 \text{ V}$ is discarded.

Verify: With $V_{o2} = 0.73 \text{ V} < 0.8 \text{ V}$ the ohmic condition holds.

Assumption is CORRECT!

Answers

$$V_{o2} = 0.73 \text{ V} \quad I_D = \frac{2 - 0.73}{1} = 1.27 \text{ mA}$$

2.2 Example 3 — A Design Problem

Problem

Design the circuit below — that is, determine the values of R_D and R_S — so that the transistor operates at $I_D = 0.4$ mA and $V_D = +0.5$ V. The NMOS transistor has $V_{GS,th} = 0.7$ V and $k = 1.6$ mA/V².

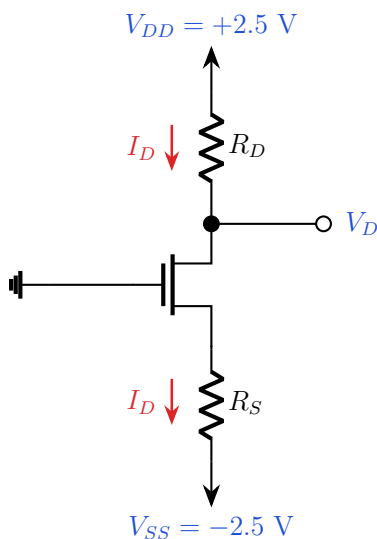


Figure 16: Circuit for Example 3. The gate is grounded, so $V_{GS} = -V_S$.

Finding R_D

The drain resistor is fixed directly by the required drain voltage and current:

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{2.5 - 0.5}{0.4} = 5 \text{ k}\Omega$$

Finding V_S

Assume: Let the MOSFET be in the **active region**: $V_{GS} > V_{GS,th}$ and $V_{DS} \geq V_{OV}$.

Solve: The gate is grounded, so $V_{GS} = V_G - V_S = -V_S$ and hence $V_{OV} = -V_S - 0.7$. The active-region equation gives

$$\begin{aligned} I_D &= k (V_{GS} - V_{GS,th})^2 = 1.6 (-V_S - 0.7)^2 = 0.4 \text{ mA} \\ (-V_S - 0.7)^2 &= 0.25 \quad \implies \quad V_S = -1.2 \text{ V} \quad \text{or} \quad -0.2 \text{ V} \end{aligned}$$

Choosing between the roots:

- $V_S = -1.2$ V gives $V_{GS} = 1.2$ V and $V_{OV} = +0.5$ V — the device conducts. **Valid.**

- $V_S = -0.2$ V gives $V_{GS} = 0.2$ V $< V_{GS,th}$, which puts the device in *cutoff* with $I_D = 0$ — contradicting $I_D = 0.4$ mA. **Discarded.**

Verify: With $V_S = -1.2$ V:

$$V_{DS} = V_D - V_S = 0.5 - (-1.2) = 1.7 \text{ V} \quad V_{OV} = -V_S - 0.7 = 0.5 \text{ V}$$

Clearly $V_{DS} \geq V_{OV}$, so the device really is in the active region.

Assumption is CORRECT!

Finding R_S

$$R_S = \frac{V_S - V_{SS}}{I_D} = \frac{-1.2 - (-2.5)}{0.4} = \frac{1.3}{0.4} = 3.25 \text{ k}\Omega$$

Answers

$$R_D = 5 \text{ k}\Omega \quad R_S = 3.25 \text{ k}\Omega \quad (V_S = -1.2 \text{ V})$$

2.3 Example 4 — A Diode-Connected MOSFET

Problem

For the circuit below, find the value of R that results in $V_D = 0.8$ V. The MOSFET has $V_{GS,th} = 0.5$ V, $k'_n = \mu_n C_{ox} = 0.4$ mA/V² and $\frac{W}{L} = \frac{0.72 \mu\text{m}}{0.18 \mu\text{m}}$.

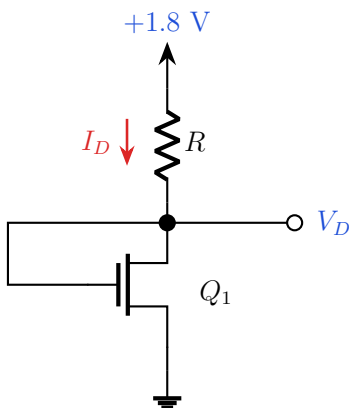


Figure 17: Circuit for Example 4. The gate is tied to the drain — a *diode-connected* MOSFET.

First collect the construction parameter:

$$k = \frac{1}{2} k'_n \frac{W}{L} = \frac{1}{2} \times 0.4 \times \frac{0.72}{0.18} = \frac{1}{2} \times 0.4 \times 4 = 0.8 \text{ mA/V}^2$$

Assume: The MOSFET *must* be in the **active region**. This one needs no guessing — it can be proved from the connection. Because the gate is tied to the drain,

$$V_D = V_G \implies V_D - V_S = V_G - V_S \implies V_{DS} = V_{GS} \geq V_{GS} - V_{GS,\text{th}}$$

which is precisely the active-region condition (it holds for any $V_{GS,\text{th}} > 0$).

A useful shortcut

A **diode-connected** MOSFET (gate shorted to drain) is *always* in the active region whenever it conducts, since $V_{DS} = V_{GS} > V_{GS} - V_{GS,\text{th}}$ automatically. No assumption or verification step is needed.

Solve: The overdrive voltage and the drain current follow immediately:

$$V_{OV} = V_{GS} - V_{GS,\text{th}} = V_{DS} - V_{GS,\text{th}} = 0.8 - 0.5 = 0.3 \text{ V}$$

$$I_D = k V_{OV}^2 = 0.8 \times (0.3)^2 = 0.072 \text{ mA}$$

The resistor then carries this current with the remaining supply voltage across it:

$$R = \frac{1.8 - V_D}{I_D} = \frac{1.8 - 0.8}{0.072} = 13.89 \text{ k}\Omega$$

Answers

$$k = 0.8 \text{ mA/V}^2 \quad I_D = 0.072 \text{ mA} \quad R = 13.89 \text{ k}\Omega$$

2.4 Example 5 — Source Degeneration with a Gate Divider

Problem

Analyse the circuit below to determine the voltages at all nodes and the currents through all branches. Let $V_{GS,th} = 1$ V and $k = 0.5$ mA/V². Neglect the channel-length modulation effect (i.e. assume $\lambda = 0$).

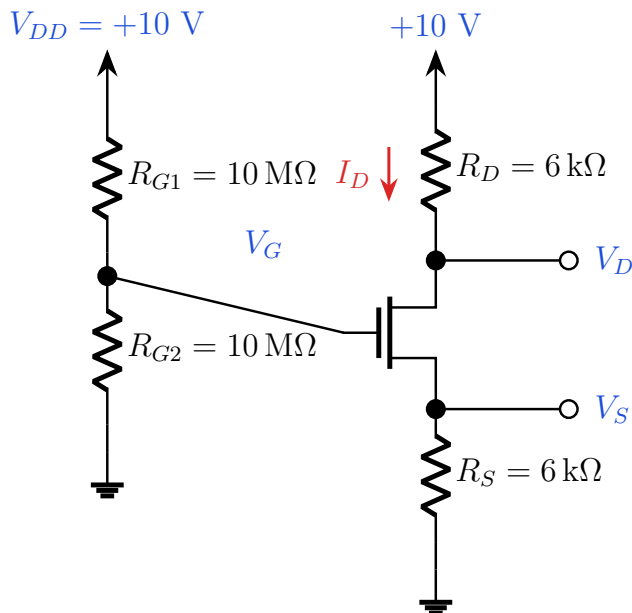


Figure 18: Circuit for Example 5. The two 10 MΩ resistors form a gate divider; the source resistor R_S provides *source degeneration*.

Because the gate draws no current, the divider is unloaded and sets

$$V_G = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 10 \times \frac{10}{10 + 10} = 5 \text{ V}$$

Assume: Let the MOSFET be in the **active region**: $V_{GS} > V_{GS,th}$ and $V_{DS} \geq V_{OV}$.

Solve: The source resistor makes $V_S = I_D R_S = 6 I_D$, so $V_{GS} = V_G - V_S = 5 - V_S$. Writing the current two ways — through R_S and from the active-region equation:

$$I_D = \frac{V_S}{6} \quad I_D = k (V_{GS} - V_{GS,th})^2 = 0.5 (4 - V_S)^2$$

Equating them,

$$\frac{V_S}{6} = \frac{1}{2} (4 - V_S)^2 \quad \Rightarrow \quad 3V_S^2 - 25V_S + 48 = 0 \quad \Rightarrow \quad V_S = 5.33 \text{ V or } 3 \text{ V}$$

Choosing between the roots:

- $V_S = 3\text{ V}$ gives $V_{GS} = 2\text{ V}$ and $V_{OV} = +1\text{ V}$ — the device conducts. **Valid.**
- $V_S = 5.33\text{ V}$ gives $V_{GS} = -0.33\text{ V} < V_{GS,\text{th}}$, i.e. cutoff. **Discarded.**

Verify: With $V_S = 3\text{ V}$:

$$I_D = \frac{V_S}{6} = \frac{3}{6} = 0.5\text{ mA}$$

$$V_D = 10 - 6I_D = 10 - 3 = 7\text{ V}$$

$$V_{DS} = V_D - V_S = 7 - 3 = 4\text{ V}$$

$$V_{OV} = V_{GS} - V_{GS,\text{th}} = 2 - 1 = 1\text{ V}$$

The active region requires $V_{DS} \geq V_{OV}$, and $4\text{ V} \geq 1\text{ V}$ holds.

Assumption is CORRECT!

Answers

$$V_G = 5\text{ V} \quad V_S = 3\text{ V} \quad V_D = 7\text{ V} \quad I_D = 0.5\text{ mA}$$

2.5 Example 6 — Sizing Two Diode-Connected Transistors

Problem

The NMOS transistors in the circuit below have $V_{GS,\text{th}} = 0.5\text{ V}$, $\mu_n C_{ox} (= k'_n) = 250\text{ }\mu\text{A/V}^2$, $\lambda = 0$ and $L_1 = L_2 = 0.25\text{ }\mu\text{m}$. Find the required gate width of each of Q_1 and Q_2 , and the value of R , to obtain the voltage and current values indicated.

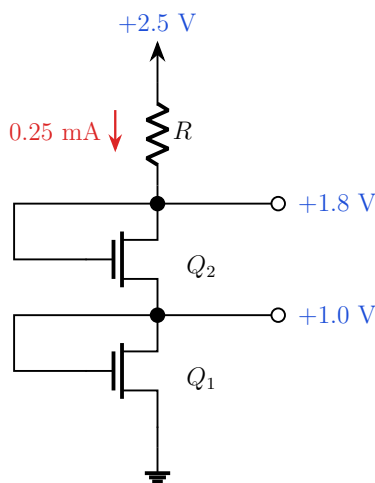


Figure 19: Circuit for Example 6. Both transistors are diode-connected, so both are in the active region.

Solve: Both transistors are diode-connected, so by the shortcut established in Example 4 **both are in the active region** — no assumption is needed. Read the gate-source voltage of each from the

node voltages:

$$\begin{aligned} Q_1 : \quad V_{GS1} - V_{GS,th} &= 1.0 - 0.5 = 0.5 \text{ V} \\ Q_2 : \quad V_{GS2} - V_{GS,th} &= (1.8 - 1.0) - 0.5 = 0.8 - 0.5 = 0.3 \text{ V} \end{aligned}$$

The same current $I_D = 0.25 \text{ mA}$ flows through both, so each must satisfy

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{GS,th})^2 = 0.25 \text{ mA}$$

With $\mu_n C_{ox} = 0.25 \text{ mA/V}^2$ and $L = 0.25 \text{ }\mu\text{m}$:

$$\begin{aligned} Q_1 : \quad \frac{0.25}{2} \cdot \frac{W_1}{0.25} (0.5)^2 &= 0.25 & \implies & & W_1 &= 2 \text{ }\mu\text{m} \\ Q_2 : \quad \frac{0.25}{2} \cdot \frac{W_2}{0.25} (0.3)^2 &= 0.25 & \implies & & W_2 &= 5.556 \text{ }\mu\text{m} \end{aligned}$$

Note that Q_2 must be substantially *wider* than Q_1 : it has to pass the same current with a smaller overdrive voltage, and $I_D \propto W V_{OV}^2$.

Finally, the resistor carries the same 0.25 mA with the remaining supply voltage across it:

$$R = \frac{2.5 - 1.8}{I_D} = \frac{0.7}{0.25} = 2.8 \text{ k}\Omega$$

Answers

$$W_1 = 2 \text{ }\mu\text{m} \quad W_2 = 5.556 \text{ }\mu\text{m} \quad R = 2.8 \text{ k}\Omega$$

3 Textbook Sample Problems

The textbook closes its Section 21 with four solved sample problems, worked here in the same style as the previous examples. Two of them concern the **depletion-type** MOSFET, so the handful of formulas those problems need is collected first.

3.1 Two New Quantities: Transconductance and Channel Resistance

Transconductance g_m and drain-source resistance R_{DS}

- The **transconductance**

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS} \text{ fixed}}$$

measures how strongly the gate voltage steers the drain current: a change δV_{GS} at the gate produces the change $\delta I_D = g_m \delta V_{GS}$ in the drain current.

- Units: mho (= siemens; 1 mho = 1 A/V). Datasheets usually quote μmhos .
- The corresponding small-signal **drain-source channel resistance** is $R_{DS} = 1/g_m$.
- For the enhancement-type NMOS in the active region, differentiating $I_D = k(V_{GS} - V_{GS,th})^2$

gives

$$g_m = 2k(V_{GS} - V_{GS,th}) = 2kV_{OV} = 2\sqrt{kI_D}$$

— the transconductance grows with the overdrive voltage.

3.2 Depletion-Type Formulas (a Preview)

The depletion-type MOSFET is *normally ON*: at $V_{GS} = 0$ it already conducts. Its datasheet is therefore built on different quantities:

Depletion-type datasheet quantities

I_{DSS} **drain current for zero bias** — the current at $V_{GS} = 0$
(gate shorted to source)

$V_{GS,off}$ **cutoff (pinch-off) voltage** — the V_{GS} that shuts the
channel (< 0 for n-channel)

$V_{GS,off}$ is the depletion counterpart of the threshold voltage $V_{GS,th}$: each marks the gate-source voltage at which the channel just disappears.

Depletion-type NMOS — active region

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS,off}}\right)^2 \quad g_{m0} = \left|\frac{2I_{DSS}}{V_{GS,off}}\right| \quad g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_{GS,off}}\right) = g_{m0} \sqrt{\frac{I_D}{I_{DSS}}}$$

g_{m0} is the transconductance at zero bias ($V_{GS} = 0$). The formulas hold for any V_{GS} above cutoff — negative (channel partly depleted) *or* positive (channel enhanced beyond I_{DSS}).

The device is in the active region when $V_{DS} \geq V_{GS} - V_{GS,off}$ — the exact analogue of the enhancement condition $V_{DS} \geq V_{OV}$, with $V_{GS} - V_{GS,off}$ playing the role of the overdrive voltage.

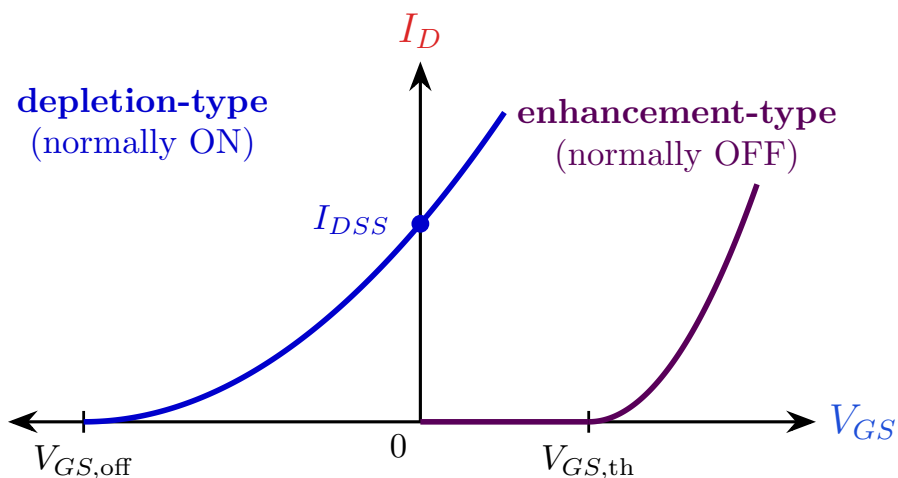


Figure 20: Transfer characteristics (I_D versus V_{GS}) of the two n-channel MOSFET kinds. The depletion-type device conducts I_{DSS} at $V_{GS} = 0$ and conducts even more for $V_{GS} > 0$; the enhancement-type device stays off until $V_{GS,th}$.

The full depletion-type section — output characteristics, ohmic-region formula and further worked examples — is still to be added separately; the formulas above are just what the textbook problems require.

3.3 Example 7 — Depletion-Type Parameters (Textbook Problem 21.1)

Problem

An n-channel depletion-type MOSFET has an $I_{DSS} = 10$ mA and a $V_{GS,off} = -4$ V. Find the values of I_D , g_m and R_{DS} when $V_{GS} = -2$ V and when $V_{GS} = +1$ V. Assume that the MOSFET is in the active region.

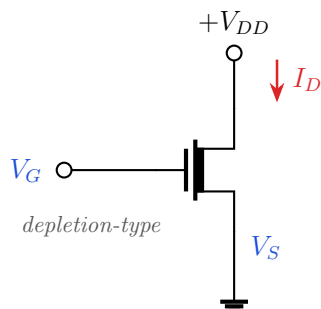


Figure 21: Bias arrangement for Example 7. The source is grounded, so $V_{GS} = V_G$.

Assume: The problem itself instructs us to take the **active region** (no V_{DS} is specified, so there is nothing to verify — the assumption is part of the given data).

Solve: At $V_{GS} = -2$ V the active-region equation gives

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS,\text{off}}} \right)^2 = 10 \left(1 - \frac{-2}{-4} \right)^2 = 10 \times (0.5)^2 = 2.5 \text{ mA}$$

The transconductance needs g_{m0} first:

$$g_{m0} = \left| \frac{2I_{DSS}}{V_{GS,\text{off}}} \right| = \frac{2 \times 10 \text{ mA}}{4 \text{ V}} = 5000 \text{ } \mu\text{mho}$$

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_{GS,\text{off}}} \right) = 5000 \times 0.5 = 2500 \text{ } \mu\text{mho} \quad R_{DS} = \frac{1}{g_m} = \frac{1}{2500 \text{ } \mu\text{mho}} = 400 \text{ } \Omega$$

Repeating for $V_{GS} = +1$ V — perfectly legal for a depletion device, whose channel is then *enhanced* beyond I_{DSS} (see the transfer curve of Fig. 20):

$$I_D = 10 \left(1 - \frac{1}{-4} \right)^2 = 10 \times (1.25)^2 = 15.6 \text{ mA} \quad g_m = 5000 \times 1.25 = 6250 \text{ } \mu\text{mho} \quad R_{DS} = 160 \text{ } \Omega$$

Answers

	I_D	g_m	R_{DS}
$V_{GS} = -2 \text{ V} :$	2.5 mA	2500 μmho	400 Ω
$V_{GS} = +1 \text{ V} :$	15.6 mA	6250 μmho	160 Ω

3.4 Example 8 — Enhancement-Type Parameters (Textbook Problem 21.2)

Problem

An n-channel enhancement-type MOSFET has a threshold voltage $V_{GS,\text{th}} = +2$ V and carries $I_D = 12$ mA when $V_{GS} = +4$ V. Find the parameters k , g_m and R_{DS} . Assume that the MOSFET is in the active region.

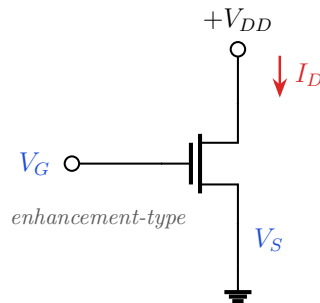


Figure 22: Bias arrangement for Example 8. The source is grounded, so $V_{GS} = V_G$.

Assume: Active region, as instructed by the problem.

Solve: Rearranging the active-region equation for the construction parameter:

$$k = \frac{I_D}{(V_{GS} - V_{GS,th})^2} = \frac{12 \text{ mA}}{(4 - 2)^2} = 3 \text{ mA/V}^2 = 3000 \text{ } \mu\text{mho/V}$$

Then

$$g_m = 2k (V_{GS} - V_{GS,th}) = 2 \times (3 \text{ mA/V}^2) \times (2 \text{ V}) = 12000 \text{ } \mu\text{mho}$$

(the alternative form gives the same: $g_m = 2\sqrt{kI_D} = 2\sqrt{3 \times 12} = 12 \text{ mA/V}$), and

$$R_{DS} = \frac{1}{g_m} = \frac{1}{0.012 \text{ mho}} \approx 83 \text{ } \Omega$$

Answers

$$k = 3 \text{ mA/V}^2 \quad g_m = 12000 \text{ } \mu\text{mho} \quad R_{DS} \approx 83 \text{ } \Omega$$

3.5 The Gain of the Common-Source Stage

The last two problems ask for the *gain* of an amplifier stage. One new idea is enough:

Where Gain = $g_m R_D$ comes from

In the circuits below, a small wiggle δV_{GS} of the input rides on top of the DC bias. It perturbs the drain current by $\delta I_D = g_m \delta V_{GS}$, and that perturbation flows through R_D , moving the output by

$$\delta V_{out} = -R_D \delta I_D = -g_m R_D \delta V_{GS} \quad \implies \quad \text{Gain} = \frac{V_{out}}{V_{in}} = g_m R_D \text{ (in magnitude)}$$

The minus sign says the stage *inverts*: raising the gate voltage increases I_D , which drags the drain voltage *down* through R_D . The book quotes the magnitude.

3.6 Example 9 — A Depletion-Type Amplifier Stage (Textbook Problem 21.3)

Problem

In the n-channel depletion-type MOSFET circuit below, $I_{DSS} = 10 \text{ mA}$, $V_{GS,off} = -4 \text{ V}$, $R_D = 1 \text{ k}\Omega$ and $V_{DD} = +20 \text{ V}$. Find V_D and the gain V_{out}/V_{in} .

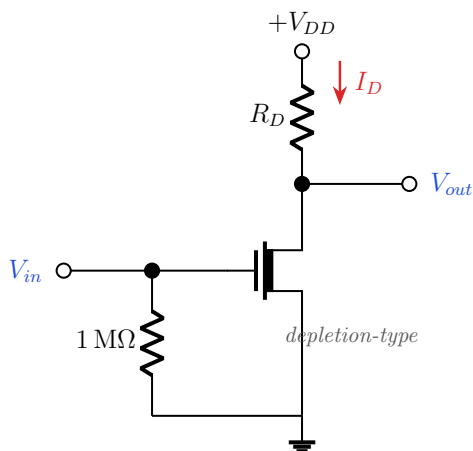


Figure 23: Circuit for Example 9. The $1 \text{ M}\Omega$ resistor gives the gate a DC path to ground.

A word about the $1 \text{ M}\Omega$ resistor: it is a *self-biasing* resistor, used to compensate leakage currents and other parameters that can lead to MOSFET instability. Because the gate leakage current is tiny (typically nA or pA), the voltage drop across it is negligible — so with no input applied, $V_{GS} = 0$.

Assume: Let the MOSFET be in the **active region**.

Solve: With $V_{GS} = 0$ the device carries exactly its zero-bias current:

$$I_D = I_{DSS} = 10 \text{ mA}$$

KVL down the drain branch (the source is grounded, so $V_{DS} = V_D$):

$$V_D = V_{DD} - I_D R_D = 20 - (10 \text{ mA})(1 \text{ k}\Omega) = 10 \text{ V}$$

The gain uses the transconductance at this operating point, and at $V_{GS} = 0$ that is precisely g_{m0} :

$$g_{m0} = \left| \frac{2I_{DSS}}{V_{GS,off}} \right| = \frac{2 \times 10 \text{ mA}}{4 \text{ V}} = 5000 \text{ }\mu\text{mho} \quad \text{Gain} = g_{m0} R_D = (5 \text{ mA/V})(1 \text{ k}\Omega) = 5$$

Verify: For the depletion device the active region requires $V_{DS} \geq V_{GS} - V_{GS,off}$. Here

$$V_{DS} = 10 \text{ V} \quad \text{and} \quad V_{GS} - V_{GS,off} = 0 - (-4) = 4 \text{ V}$$

so the condition holds comfortably.

Assumption is **CORRECT!**

Answers

$$V_D = 10 \text{ V} \quad \text{Gain} = g_{m0} R_D = 5$$

3.7 Example 10 — An Enhancement-Type Amplifier Stage (Textbook Problem 21.4)

Problem

In the n-channel enhancement-type MOSFET circuit below, if $k = 1000 \mu\text{mho}/\text{V}^2$, $V_{DD} = 20 \text{ V}$, $V_{GS,\text{th}} = 2 \text{ V}$ and $V_{GS} = 5 \text{ V}$, what should the resistance of R_D be to center V_D at 10 V ? Also, what is the gain for this circuit?

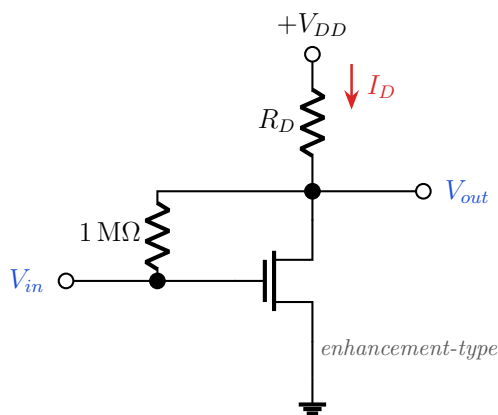


Figure 24: Circuit for Example 10. The $1 \text{ M}\Omega$ resistor now runs from gate to *drain*.

The $1 \text{ M}\Omega$ resistor plays the same stabilising role as in Example 9, but it now runs from gate to *drain*: an enhancement device must have its gate held *above* threshold, and the drain is a convenient positive node. Since the gate draws no current, the current it carries,

$$\frac{V_D - V_G}{1 \text{ M}\Omega} = \frac{10 - 5}{1 \text{ M}\Omega} = 5 \mu\text{A},$$

is negligible beside the drain current — it disturbs neither node.

Assume: Let the MOSFET be in the **active region**.

Solve: The drain current is fixed by the given bias:

$$I_D = k (V_{GS} - V_{GS,\text{th}})^2 = (1000 \mu\text{mho}/\text{V}^2) \times (5 - 2)^2 = 9 \text{ mA}$$

Ohm's law on R_D then centres the drain at 10 V :

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{20 - 10}{9 \text{ mA}} = 1111.1 \Omega \approx 1.11 \text{ k}\Omega$$

For the gain, first the transconductance at this bias:

$$g_m = 2k(V_{GS} - V_{GS,th}) = 2 \times (1000 \mu\text{mho}/\text{V}^2) \times (3 \text{ V}) = 6000 \mu\text{mho}$$

$$\text{Gain} = g_m R_D = (6 \text{ mA/V})(1111.1 \Omega) \approx 6.7$$

Verify: $V_{DS} = V_D = 10 \text{ V}$ and $V_{OV} = V_{GS} - V_{GS,th} = 3 \text{ V}$; since $V_{DS} \geq V_{OV}$, the device really is in the active region.

Assumption is CORRECT!

Answers

$$I_D = 9 \text{ mA} \quad R_D \approx 1.11 \text{ k}\Omega \quad \text{Gain} = g_m R_D \approx 6.7$$